Compounds and molecules

ATOMS JOIN TOGETHER TO FORM COMPOUNDS AND MOLECULES.

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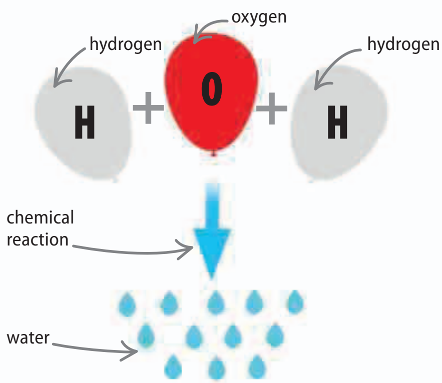
Few elements exist naturally in their pure form. Gold is one example. Most other elements form compounds, when their atoms bond with those of other elements.

What is a compound?

Almost all everyday items are made up of chemical compounds, from the water coming out of the tap to the minerals in bricks and stones to the substances in the human body. A compound is a single substance made up of the atoms of two or more elements, which are chemically connected or bonded. This differentiates a compound from a mixture, which is made up of two or more separate substances.



◁ **Fixed ratio**
A compound's constituent elements have a fixed ratio. Water (H_2O) has two parts of hydrogen (H) for every part of oxygen (O).

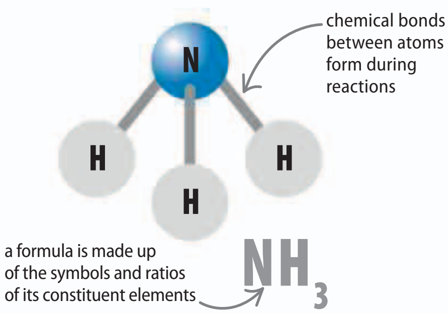
◁ **Chemical reactions**
A compound can only be formed when elements, or other compounds, react with each other.

◁ **Different properties**
A compound's properties are different to those of its constituent elements. Water is a liquid, for example, that is made up of two gases.

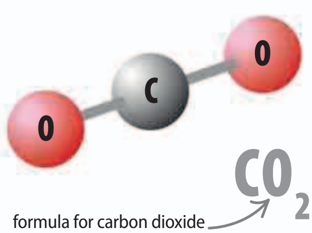
Molecules

A molecule is the smallest unit of a compound. Breaking the molecule down into simpler constituents would result in the compound ceasing to exist. The atoms in a molecule are connected by chemical bonds. The arrangement and strength of the bonds gives the molecule a certain shape.

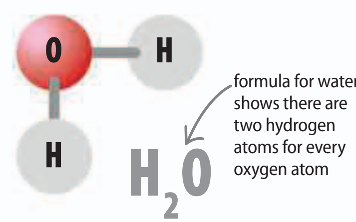
At **very high pressures,** oxygen molecules transform into an eight-atom version that is bright red.



△ **Ammonia**
The molecules of this compound have a single nitrogen atom (N) bonded to three hydrogen atoms (H). Together they form a tetrahedron.



△ **Carbon dioxide**
As its name suggests, this compound has one carbon atom (C) bonded to two oxygen atoms (O). The three atoms form a straight molecule.



△ **Water**
Common compounds such as water have nonscientific names. Others, such as carbon dioxide, are named according to the elements they contain.